	Pimpri Chinchwad Education Trust's **Pimpri Chinchwad College of Engineering** Department of Information Technology
Assignment 7	

NAME -	Viraj Pathare
PRN -	124B1F149
CLASS -	SY - IT
DIV -	A

Aim: Design and Develop Queries using CRUD operations. (Use CRUD operations, SAVE method, logical operators, joins etc.).

Theory:

Q1. What are CRUD operations in MongoDB?

Answer:
CRUD stands for Create, Read, Update, and Delete operations. These are the basic operations used to manage data in MongoDB.

- Create → insert data (`insertOne`, `insertMany`)
- Read → retrieve data (`find`)
- Update → modify data (`updateOne`, `updateMany`)
- Delete → remove data (`deleteOne`, `deleteMany`)

Q2. What is the use of logical operators in MongoDB?

Answer:
Logical operators in MongoDB are used to combine multiple conditions in a query.

Common operators include:

- `$and` → all conditions must be true
- `$or` → at least one condition must be true
- `$not` → negates a condition

They help in filtering data more efficiently based on multiple criteria.

Q3. What is $lookup in MongoDB?

Answer:

`$lookup` is used in MongoDB to perform join operations between collections. It combines data from different collections based on a common field, similar to SQL joins. It is mainly used in the aggregation framework to retrieve related data.

Implementation

Step 1: Create Database & Collection

Step 2: INSERT (Create Operation)

```
mongosh mongodb://127.0.0.

test> use CompanyDB
switched to db CompanyDB
CompanyDB>
(To exit, press Ctrl+C again or Ctrl+D or type .exit)
CompanyDB> db.createCollection("employees")
{ ok: 1 }
CompanyDB> db.employees.insertMany([
| { eid: 1, name: "Amit", salary: 50000, department: "IT" },
| { eid: 2, name: "Neha", salary: 40000, department: "HR" },
| { eid: 3, name: "Raj", salary: 30000, department: "Finance" },
| { eid: 4, name: "Simran", salary: 45000, department: "IT" }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69de7a4c9b2d58ddc63682d1'),
    '1': ObjectId('69de7a4c9b2d58ddc63682d2'),
    '2': ObjectId('69de7a4c9b2d58ddc63682d3'),
    '3': ObjectId('69de7a4c9b2d58ddc63682d4')
  }
}
```

Step 3: READ (Retrieve Data)

```
mongosh mongodb://127.0.0.
    '2': ObjectId('69de7a4c9b2d58ddc63682d3'),
    '3': ObjectId('69de7a4c9b2d58ddc63682d4')
  }
}
CompanyDB> // Display all employees
| db.employees.find()
|
| // Employees with salary > 40000
| db.employees.find({ salary: { $gt: 40000 } })
|
| // Using logical operator (AND)
| db.employees.find({
| $and: [
| { department: "IT" },
| { salary: { $gt: 40000 } }
| ]
| })
[
  {
    _id: ObjectId('69de7a4c9b2d58ddc63682d1'),
    eid: 1,
    name: 'Amit',
    salary: 50000,
    department: 'IT'
  },
  {
    _id: ObjectId('69de7a4c9b2d58ddc63682d4'),
    eid: 4,
    name: 'Simran',
    salary: 45000,
    department: 'IT'
  }
]
```

Step 4: UPDATE Operation

```
mongosh mongodb://127.0.0.
    department: 'IT'
  }
]
CompanyDB> // Increase salary of Amit
| db.employees.updateOne(
| { name: "Amit" },
| { $set: { salary: 55000 } }
| )
|
| // Increase salary of all IT employees
| db.employees.updateMany(
| { department: "IT" },
| { $inc: { salary: 2000 } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
```

Step 5: DELETE Operation

```
CompanyDB> // Delete one employee
| db.employees.deleteOne({ name: "Raj" })
|
| // Delete employees with salary < 35000
| db.employees.deleteMany({ salary: { $lt: 35000 } })
{ acknowledged: true, deletedCount: 0 }
CompanyDB> db.employees.updateOne(
| { eid: 5 },
| { $set: { name: "Karan", salary: 38000, department: "HR" } },
| { upsert: true }
| )
{
  acknowledged: true,
  insertedId: ObjectId('69de7a764f4893c33759ccd8'),
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 1
}
```

Step 6: SAVE Method (Upsert Concept)

Step 7: JOIN (Using $lookup)

```
mongosh mongodb://127.0.0.
  modifiedCount: 0,
  upsertedCount: 1
}
CompanyDB> // Create another collection
| db.projects.insertMany([
| { pid: 101, pname: "Bank Management", eid: 1 },
| { pid: 102, pname: "E-Commerce", eid: 2 }
| ])
|
| // Join employees with projects
| db.employees.aggregate([
| {
| $lookup: {
| from: "projects",
| localField: "eid",
| foreignField: "eid",
| as: "project_details"
| }
| }
| ])
[
  {
    _id: ObjectId('69de7a4c9b2d58ddc63682d1'),
    eid: 1,
    name: 'Amit',
    salary: 57000,
    department: 'IT',
    project_details: [
      {
        _id: ObjectId('69de7a869b2d58ddc63682d5'),
        pid: 101,
        pname: 'Bank Management',
        eid: 1
      }
    ]
  },
  {
    _id: ObjectId('69de7a4c9b2d58ddc63682d2'),
    eid: 2,
    name: 'Neha',
    salary: 48000,
    department: 'HR',
    project_details: [
      {
```